

## STARD 2015 checklist for reporting of studies of diagnostic accuracy

| Section & Topic   | No  | Item                                                                                                                                                   | Reported on page # |
|-------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| TITLE OR ABSTRACT |     |                                                                                                                                                        |                    |
|                   | 1   | Identification as a study of diagnostic accuracy using at least one measure of accuracy (such as sensitivity, specificity, predictive values, or AUC)  | 1                  |
| ABSTRACT          |     |                                                                                                                                                        |                    |
|                   | 2   | Structured summary of study design, methods, results, and conclusions (for specific guidance, see STARD for Abstracts)                                 | ⚠                  |
| INTRODUCTION      |     |                                                                                                                                                        |                    |
|                   | 3   | Scientific and clinical background, including the intended use and clinical role of the index test                                                     | ⚠                  |
|                   | 4   | Study objectives and hypotheses                                                                                                                        | ⚠                  |
| METHODS           |     |                                                                                                                                                        |                    |
| Study design      | 5   | Whether data collection was planned before the index test and reference standard were performed (prospective study) or after (retrospective study)     | 1                  |
| Participants      | 6   | Eligibility criteria                                                                                                                                   | 1                  |
|                   | 7   | On what basis potentially eligible participants were identified (such as symptoms, results from previous tests, inclusion in registry)                 | ⚠                  |
|                   | 8   | Where and when potentially eligible participants were identified (setting, location and dates)                                                         | ⚠                  |
|                   | 9   | Whether participants formed a consecutive, random or convenience series                                                                                | ⚠                  |
| Test methods      | 10a | Index test, in sufficient detail to allow replication                                                                                                  | 1                  |
|                   | 10b | Reference standard, in sufficient detail to allow replication                                                                                          | 1                  |
|                   | 11  | Rationale for choosing the reference standard (if alternatives exist)                                                                                  | ⚠                  |
|                   | 12a | Definition of and rationale for test positivity cut-offs or result categories of the index test, distinguishing pre-specified from exploratory         | ⚠                  |
|                   | 12b | Definition of and rationale for test positivity cut-offs or result categories of the reference standard, distinguishing pre-specified from exploratory | ⚠                  |
|                   | 13a | Whether clinical information and reference standard results were available to the performers/readers of the index test                                 | ⚠                  |
|                   | 13b | Whether clinical information and index test results were available to the assessors of the reference standard                                          | ⚠                  |
| Analysis          | 14  | Methods for estimating or comparing measures of diagnostic accuracy                                                                                    | ⚠                  |
|                   | 15  | How indeterminate index test or reference standard results were handled                                                                                | ⚠                  |
|                   | 16  | How missing data on the index test and reference standard were handled                                                                                 | ⚠                  |
|                   | 17  | Any analyses of variability in diagnostic accuracy, distinguishing pre-specified from exploratory                                                      | ⚠                  |
|                   | 18  | Intended sample size and how it was determined                                                                                                         | ⚠                  |

| Section & Topic          | No  | Item                                                                                                        | Reported on page # |
|--------------------------|-----|-------------------------------------------------------------------------------------------------------------|--------------------|
| <b>RESULTS</b>           |     |                                                                                                             |                    |
| Participants             | 19  | Flow of participants, using a diagram                                                                       | 1                  |
|                          | 20  | Baseline demographic and clinical characteristics of participants                                           | 1                  |
|                          | 21a | Distribution of severity of disease in those with the target condition                                      | ⚠                  |
|                          | 21b | Distribution of alternative diagnoses in those without the target condition                                 | ⚠                  |
|                          | 22  | Time interval and any clinical interventions between index test and reference standard                      | ⚠                  |
| Test results             | 23  | Cross tabulation of the index test results (or their distribution) by the results of the reference standard | ⚠                  |
|                          | 24  | Estimates of diagnostic accuracy and their precision (such as 95% confidence intervals)                     | ⚠                  |
|                          | 25  | Any adverse events from performing the index test or the reference standard                                 | ⚠                  |
| <b>DISCUSSION</b>        |     |                                                                                                             |                    |
|                          | 26  | Study limitations, including sources of potential bias, statistical uncertainty, and generalisability       | 1                  |
|                          | 27  | Implications for practice, including the intended use and clinical role of the index test                   | ⚠                  |
| <b>OTHER INFORMATION</b> |     |                                                                                                             |                    |
|                          | 28  | Registration number and name of registry                                                                    | ⚠                  |
|                          | 29  | Where the full study protocol can be accessed                                                               | ⚠                  |
|                          | 30  | Sources of funding and other support; role of funders                                                       | ⚠                  |

### Diagnostics

⚠ item 2: not covered (no check(), no na())

⚠ item 3: not covered (no check(), no na())

⚠ item 4: not covered (no check(), no na())

⚠ item 7: not covered (no check(), no na())

⚠ item 8: not covered (no check(), no na())

⚠ item 9: not covered (no check(), no na())

⚠ item 11: not covered (no check(), no na())

⚠ item 12a: not covered (no check(), no na())

⚠ item 12b: not covered (no check(), no na())

⚠ item 13a: not covered (no check(), no na())

⚠ item 13b: not covered (no check(), no na())

⚠ item 14: not covered (no check(), no na())

⚠ item 15: not covered (no check(), no na())

⚠ item 16: not covered (no check(), no na())

⚠ item 17: not covered (no check(), no na())

⚠ item 18: not covered (no check(), no na())

⚠ item 21a: not covered (no check(), no na())

⚠ item 21b: not covered (no check(), no na())

⚠ item 22: not covered (no check(), no na())

⚠ item 23: not covered (no check(), no na())

⚠ item 24: not covered (no check(), no na())

⚠ item 25: not covered (no check(), no na())

⚠ item 27: not covered (no check(), no na())

⚠ item 28: not covered (no check(), no na())

⚠ item 29: not covered (no check(), no na())

⚠ item 30: not covered (no check(), no na())

**AIM** — STARD stands for “Standards for Reporting Diagnostic accuracy studies”. This list of items was developed to contribute to the completeness and transparency of reporting of diagnostic accuracy studies. Authors can use the list to write informative study reports. Editors and peer-reviewers can use it to evaluate whether the information has been included in manuscripts submitted for publication.

**Explanation** — A diagnostic accuracy study evaluates the ability of one or more medical tests to correctly classify study participants as having a target condition. The test whose accuracy is evaluated is called the index test; the reference standard is the best available method for establishing the presence or absence of the target condition.

**Development** — This STARD list was released in 2015. The 30 items were identified by an international expert group of methodologists, researchers, and editors, and represent an update of the first version, published in 2003. More information can be found at [www.equator-network.org/reporting-guidelines/stard](http://www.equator-network.org/reporting-guidelines/stard).